Protocol Addendum A1: timing, preprocessing and timing-aware tests

Consolidated frozen protocol for steps 2–6. Version 1.2. Status: **FROZEN** on 30 Oct 2026. Production runs start only after Stage 0 passes and its projected runtime is approved.

Machine-readable part: Simulation_Addendum_A1_v1.2_FROZEN.json, SHA-256 07e896f9bc70eb1356efbfcbfce08b0e6e06d470c28485f5ed592d66d0bffe4b

Parent configuration 0473d4830407f1fde68988d32fb1d2ad14537b815c0fd34492b8391d084950d3 (Phase 2); grandparent 9a245eb092a47bc5e2851f0b681ec439c1a72ab34ed2fe3dedf63b9b29bff482 (Phase 1). Every parent value is unchanged; A1 only adds.

Change in v1.2: the HIGH label is split. HIGH_NOT_EXCLUDED (upper limit above 7.5%) means false-alarm control is **not established**; only HIGH (lower limit above 7.5%) permits “at the cost of excess false alarms” (Sections 5b, 6.2, 6.5, 7). **Changes in v1.1:** (1) the calibration of **every** compared test, not only ou_ss, now governs each comparative claim, with mechanical labels and a pre-specified size-adjusted analysis (Section 5b; 6.1, 6.2, 7); (2) E2FA n = 1,000 is the target; a reduction to 500 needs recorded approval and is strictly descriptive (6.7); (3) every detection change is classified four ways, and only GENUINE_RECOVERY counts as recovery (5b, 6.5, 6.7, 7).

1. Scope and boundaries

- **In scope:** step 2 (E2 continuous-time method and its false alarms), step 3 (slow-process false alarms), step 4 (standard vs timing-aware tests across scenarios), step 5 (preprocessing with standard tests), step 6 (other decline shapes and their false alarms).
- **Separate and unchanged:** the earlier E1/E3 runs (frozen exploratory blocks, run with ewsv322/explore.py, SHA-256 7161cf94...9d5c). Their configuration is not touched and no requirement is added to them.
- **Not simulation, handled outside A1:** step 7 (MIMIC-IV real-data anchor) gets its own analysis plan; steps 8–9 (literature, statistical write-up) are writing tasks.

- **Principle:** an experiment is included only if it answers a specific open question listed in Section 11. Criticisms that need explanation or literature, not data, get no new run.
- **Never:** overwrite completed Phase 2 outputs; change a null method, margin or decision rule after seeing any A1 result; tune a model on the informative arm.

2. The three scientific questions and their estimands

All estimands are paired within patient (same latent path), detection proportions in percentage points (pp), “not evaluable” counted as not detected, as frozen.

Question	Estimand (per indicator: variance, lag-1 AC)	Main comparison groups
Q1 Does timing change detection at matched count?	det(informative_reduced_N) − det(regular_N), index, ou_ss. Decomposed: **irregularity** = det(poisson_matched) − det(regular_N); **escalation** = det(informative_reduced_N) − det(poisson_matched).	F1 (fresh-seed replication), XS1, XR1–XR10, XS2, FOLD2
Q2 Can preprocessing recover detection?	For representation r and test n: det(inf, r, n) − det(inf, index, n) (recovery) and det(inf, r, n) − det(reg, index, ou_ss) (gap to even sampling), each read against FA(r, n).	E1/E3 (ou_ss, exploratory, frozen) + P5/P5FA (ar1_rep, shuffle_rep); E2/E2FA (continuous-time)
Q3 Does a timing-aware test do better?	det(inf, index, ou_ss) − det(inf, index, ar1) (gain); specificity: gain(inf) − gain(reg); read against FA of every test in slow and fast stable processes.	G1a–d, F1 (confirmatory), XN1, XN2, all X-blocks

3. The three tests, defined exactly

Each test generates 200 surrogate series under “no transition”, passes each through the identical analysis, and computes the plus-one Monte Carlo p-value p = (1 + #{surrogate tau ≥ observed tau, or surrogate invalid}) / 201. Detection = observed Kendall tau-a > 0 and p < 0.05. Surrogate k of (arm, analysis, test) uses stream spawn_key (K(‘surrogates’), K(‘<arm>|<analysis>|<test>’), k), entropy = seed base + patient id. All three are already implemented and hash-locked (nulls.py, kalman.py).

ou_ss: timing-aware (state-space OU at the observed timestamps). Fit to the arm's whole analysed series (t_i, y_i) the model

$X_1 \sim N(\mu, V)$; $X_{i+1} = \mu + e^{-\theta\Delta_i}(X_i - \mu) + \eta_i$, $\Delta_i = t_{i+1} - t_i$

$\eta_i \sim N(0, V(1 - e^{-2\theta\Delta_i}))$; $Y_i = X_i + \varepsilon_i$, $\varepsilon_i \sim N(0, \rho V)$

with μ = sample mean, exact Gaussian likelihood by Kalman filter, (θ, ρ) maximised on a fixed grid (θ: 32 log-spaced values 0.01–20; ρ: 0, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6, 3.2) with V profiled in closed form. Surrogates are simulated from the fitted model **at the arm's own timestamps**, so they reproduce the effect of the actual spacing.

ar1: timestamp-blind (AR(1) in observation order). φ = lag-1 least-squares coefficient of the demeaned series, clipped to [−0.99, 0.99]; s = SD(y) (ddof 0); surrogate $s_1 = s \cdot z_1$, $s_i = \varphi\, s_{i-1} + s\sqrt{(1-\varphi^2)}\, z_i$, plus the series mean. Ignores the time between observations (the default in common EWS software, e.g. Dakos et al. 2012).

shuffle: timestamp-blind (random permutation) of the observed values; destroys all serial dependence.

New for representations (step 5): ar1_rep and shuffle_rep. The analyst pipeline “regularise first, then use a standard test”: z = R(t, y) is the binned or interpolated series of length L; fit ar1 to z (same rule) or permute z; compute the window statistics directly on z* (z* is not re-binned). For index, ar1_rep = ar1 and shuffle_rep = shuffle. For fixed_time|W the representation is the raw series, so ar1_rep is the frozen ar1 (observation order, assigned to the observed timestamps). This differs from the frozen code path, where ar1 surrogates are generated on the raw series and then binned; the A1 definition matches what an analyst actually does.

4. New components

Component	Definition	Why it is needed
Model stable_slow_0.2	Stationary OU, $\theta = 0.2$ fixed, $\mu = 0$, $\sigma = 0.3$, $X(0) \sim N(0, \sigma^2/(2 \cdot 0.2))$; same measurement error SD 0.1, baseline and schedules.	Median fitted ou_ss θ in P1 transition series is 0.15–0.19. Phase 1 calibrated at $\theta = 1.2$ only, where spacing barely matters (lag correlation about 0.14). The timing effect acts in the slow regime.
Model stable_slow_0.05	As above with $\theta = 0.05$ (the value at the end of the ou_csd transition).	Bounds the slow regime from below.
Model fold_v2	Frozen fold model with (σ_f, r_{end}) chosen from the grid $\sigma_f \in \{0.05, 0.07, 0.10\} \times r_{end} \in \{0.05, 0.10\}$ by the pilot rule in Section 6.6.	Frozen fold (S3) had 11.4% tipping and 19.4% regular-arm detection: too little power to inform anything.
Arm poisson_matched	8 baseline observations, then $N_i - 8$ times i.i.d. Uniform[15, 100), sorted (homogeneous Poisson times conditional on count), stream 'pm_times'; own error stream 'err_pm'.	Exactly count-matched, state-independent random timing. Cleaner than thinning (thinned counts ranged 39–84). Defined for every schedule, including smooth. Separates irregularity from escalation.
Tests ar1_rep, shuffle_rep	Section 3.	Step 5: "fix the data" with a standard test vs "fix the test".

5. Hierarchy of evidence and multiplicity

Level	What	Inference rule
Gate	G1: ou_ss passes TOST (2.5–7.5%, 90% Clopper–Pearson inside bounds) for variance and AC in regular_N, informative_reduced_N and poisson_matched in G1a (stable_slow_0.2, m = 2, primary trigger).	Fail → no Q3 claim in the paper; H1/H2 reported as descriptive only.
Confirmatory (fresh seeds, never analysed before)	H1 and H2 in block F1 (Section 6.2).	Family-wise α 0.05 by Bonferroni: each judged on its two-sided 97.5% paired bootstrap CI.
Pre-specified secondary	Families: S-Q1 (F1 replication and decomposition, XS1, XR, XS2, FOLD2); S-Q2 (P5, E2); S-Q3 (gain and decomposition by test across X-blocks); S-FA (every false-alarm block).	95% CIs labelled unadjusted, plus Holm-adjusted exact McNemar p-values within each family. Interpretation categories use the frozen ±5 pp rule.
Exploratory	E1/E3 (frozen as exploratory, cannot be upgraded), tau-scale subgroup analyses, escalation-time cuts, anything not listed above.	Estimates and CIs only; never used as the basis of a claim.

The ±5 pp margin. Kept exactly as frozen in Phase 2 (changing it now would be outcome-driven). Pre-stated justification: (i) at baseline detection near 50%, 5 pp is a 10% relative change, the smallest change we consider practically meaningful for a screening indicator; (ii) it is close to the resolution of the design: with n = 1,000 paired patients and discordance about 0.40, the 95% CI half-width is $1.96 \cdot \sqrt{(0.40/1000)} \approx 3.9$ pp, so smaller differences cannot be classified reliably. Sensitivity: every interpretation category is also reported at ±3 and ±10 pp.

Continuous outcome. For every comparison, mean Kendall tau (variance and AC) per arm and the paired difference with bootstrap CI are reported next to detection, so attenuation is visible even where detection does not move.

Correlated blocks. X-blocks share patients and latent paths with each other and with Phase 2 by design (common random numbers). Agreement across them is reported as consistency, never as independent replication; only F1 (fresh seeds) is independent.

5b. Calibration status of every test, and how it governs claims (new in v1.1)

Every test is labelled in every null-process block, arm and indicator, mechanically from the 90% Clopper–Pearson interval [L, U] of its false-alarm rate:

Label	Rule	Meaning for claims
CALIBRATED	L ≥ 2.5% and U ≤ 7.5% (the frozen TOST)	May enter a claim at controlled false-alarm rates.
LOW	U ≤ 7.5% and L < 2.5%	Conservative: fewer false alarms than nominal. Comparisons against it are worded as such.
HIGH_NOT_EXCLUDED	U > 7.5% and L ≤ 7.5%	Excess false alarms not ruled out (but not shown): false-alarm control **not established**. Cannot support a superiority or recovery claim.
HIGH	L > 7.5%	False-alarm rate demonstrably above the upper bound. Only this label permits the wording “at the cost of excess false alarms”.

Reference blocks. Q3 (tests on the raw series): G1a (slow regime, primary) and XN2 (θ = 1.2). A test counts as CALIBRATED for a Q3 comparison only if it is CALIBRATED in both, in the arm(s) the comparison uses, for that indicator. Q2: P5FA for ar1_rep and shuffle_rep, E3 for the ou_ss representations, E2FA for ct|W30.

Size-adjusted sensitivity analysis (pre-specified). Run for every Q3 comparison, and the basis of interpretation whenever a compared test is not CALIBRATED: for each test and arm, the largest p-value threshold α* on the Monte Carlo grid {k/201} giving a false-alarm rate ≤ 5% in G1a (same test, arm and indicator) is found; F1 detection is re-classified as tau > 0 and p ≤ α*; the H1/H2 estimands and 97.5% CIs are recomputed. Repeated with XN2 as reference. Limitation stated: the bootstrap does not re-estimate α*.

Four-way classification of any detection change (Q2 recovery, and Q3 gains):

Class	Condition	May be called
NO_RECOVERY	95% CI of the change includes or lies below 0	No change / loss
IMPROVEMENT_FA_CONTROL_NOT_ESTABLISHED	CI lower bound > 0; method HIGH_NOT_EXCLUDED, or false-alarm result not available	“Detection improvement with false-alarm control not established”; never recovery
GENUINE_RECOVERY	CI lower bound > 0 and the method CALIBRATED or LOW in the matching null block (same arm, indicator)	“Recovers detection” (material if lower bound > +5)
IMPROVEMENT_AT_FALSE_ALARM_COST	CI lower bound > 0 and the method HIGH (90% lower limit > 7.5%)	“Detection improvement at the cost of excess false alarms”

6. Experiments

Each experiment lists the eight items requested: question and hypothesis; model and comparison groups; sample size and seeds; outcomes and tests; false-alarm controls; CIs and multiplicity; interpretation criteria; outputs and checks. Item 8 is common to all and is given once in Section 9.

6.1 G1a–G1d: false alarms in the slow regime (step 3)

Item	Specification
Question / hypothesis	Q3 calibration. H_G: ou_ss false-alarm rate is within 2.5–7.5% in a slow stationary process with state-dependent escalating sampling. No directional hypothesis for ar1/shuffle.
Model, groups	G1a stable_slow_0.2, m = 2 (gate); G1b stable_slow_0.2, m = 6; G1c stable_slow_0.05, m = 2; G1d stable_slow_0.05, m = 6. Step schedule, primary trigger. Arms regular_N, informative_reduced_N, poisson_matched. Tests ou_ss, ar1, shuffle. Index analysis.
n, seeds	2,000 patients per block, ids 0–1999, seed base 6,000,000 (fresh). Same ids share latent paths within a θ. Precision: TOST at n = 2,000 passes for observed 3.2–6.5%; probability of passing at a true 5% is 0.999.
Outcomes, tests	False-alarm rate per arm × test × indicator, exact TOST, 90% and 95% Clopper–Pearson; paired difference informative − regular with bootstrap CI.
CIs, multiplicity	G1a is the gate (no multiplicity: all six ou_ss cells must pass). G1b–d belong to family S-FA.
Interpretation	G1a pass → Q3 may be considered. Every test (ou_ss, ar1, shuffle) receives a mechanical calibration label in every block and arm (Section 5b). These labels, not the gate alone, govern the wording of every comparative claim. G1b–d: ou_ss CALIBRATED → claim extends to that scope; otherwise scope restriction stated.

6.2 F1: confirmatory test of the timing-aware test (step 4, primary)

Item	Specification
Question / hypothesis	Q3. **H1:** variance detection in informative_reduced_N is higher with ou_ss than with ar1. **H2:** that gain is larger under informative sampling than under even sampling. AC: no gain predicted (secondary, assessed against ±5 pp).
Model, groups	ou_csd, step, m = 2, primary trigger (P1 design). Arms regular_N, informative_reduced_N, thinned, poisson_matched. Tests ou_ss, ar1, shuffle. Index analysis.
n, seeds	2,000 patients, ids 0–1999, seed base 5,000,000: never analysed before, so the finding discovered in P1 is tested on independent data. Power: P1 gave a gain of +13.3 pp (95% CI 11.0 to 15.5) at n = 1,000; at n = 2,000 the 97.5% CI half-width is about ±1.8 pp, so power to exclude 0 (and to exclude +5) is > 0.99 if the true gain is even half the P1 value.
Estimands	H1: Δ_1 = det(inf, ou_ss) − det(inf, ar1). H2: Δ_2 = [det(inf, ou_ss) − det(inf, ar1)] − [det(reg, ou_ss) − det(reg, ar1)]. Same for shuffle in place of ar1 (secondary).
False-alarm controls	G1a (gate), XN2 (stable, θ = 1.2, all tests), XN1 (level drop, all tests).
CIs, multiplicity	H1 and H2: two-sided 97.5% paired bootstrap CIs (10,000 resamples, patient unit), Newcombe cross-check for Δ_1; Bonferroni family. Everything else in F1 is secondary (S-Q1, S-Q3).
Interpretation	Statistical rule: H1 supported if lower 97.5% bound > 0 (material if > +5); H2 supported if lower bound > 0. **Claim rule:** support is necessary, not sufficient. “Better at controlled false-alarm rates” requires ou_ss **and** the comparator to be CALIBRATED in G1a and XN2 in the arm(s) used (informative_reduced_N for H1; both arms for H2). Comparator LOW → “detects more than a conservative test”, judged on the size-adjusted analysis. Comparator HIGH_NOT_EXCLUDED → no claim at controlled false-alarm rates (comparator’s control not established); judged on the size-adjusted analysis. Comparator HIGH → tests of demonstrably unequal size; no superiority claim from the raw comparison; judged only on the size-adjusted analysis. H1 and H2 supported with all tests CALIBRATED → irregularity-specific claim. H1 only → “more powerful in general”. H1 not supported → Q3 dropped. Q1 replication in F1 by the frozen Phase 2 rule.

6.3 XS1, XR1–XR10: generality across scenarios (step 4, secondary)

Item	Specification
Question / hypothesis	Do Q1 and Q3 hold across trigger rules, escalation sizes and schedule shapes, and is the escalation component about state dependence or about any rise in sampling density?
Model, groups	Exactly the frozen S1 (independent trigger) and R1–R10 (single, k = 2.0, k = 2.5, oracle triggers; m = 1.5, 3, 4, 6; ramp; smooth). Arms regular_N, informative_reduced_N, thinned (not for smooth), poisson_matched. Tests ou_ss, ar1, shuffle. Index.
n, seeds	XS1: ids 0–999; XR: ids 0–999 (frozen R blocks had 0–499; 500–999 added for precision). Seed base 3,000,000, so existing rows must be reproduced exactly (Section 9). Paired 95% CI half-width at n = 1,000 is about ±3.9 pp.
Outcomes	By block and test: Q1 total, irregularity and escalation components; Q3 gain and specificity; tau-scale versions.
CIs, multiplicity	95% unadjusted CIs; Holm within family S-Q1 and S-Q3.
Interpretation	State dependence may be claimed only if the escalation component differs between P1-design (F1 or P1) and XS1 with a 95% CI excluding 0; otherwise the paper says “rise in sampling density”, never “state-dependent”. Dose-response in m is described, with a paired linear trend in log(m) across XR5, P1/F1, XR6–XR8 reported as secondary. Note: for the ramp (XR9) thinning by 1/m does not restore the base rate, because the rate already rises before escalation; poisson_matched is the comparator there.

6.4 XS2, XN1, XN2: decline shape and level-drop false alarms (step 6)

Item	Specification
Question / hypothesis	Separate genuine slowing from a level drop. XS2 (slowing + level drop): do Q1/Q3 hold when the ou_ss constant-mean assumption is violated? XN1 (level drop, no slowing) and XN2 (stable, θ = 1.2): false alarms of every test under state-dependent sampling.
Model, groups	Frozen S2, N1, N2 designs. XS2 arms: four. XN1/XN2 arms: regular_N, informative_reduced_N, poisson_matched. Tests ou_ss, ar1, shuffle.
n, seeds	XS2 ids 0–999; XN1, XN2 ids 0–1999; seed base 3,000,000 (existing ou_ss rows reproduced exactly).
Outcomes	XS2 as 6.3. XN1/XN2 false-alarm rates, TOST, paired informative − regular difference.
Interpretation	If XS2's Q3 gain (ou_ss vs ar1) is below 5 pp, the Q3 claim is restricted to constant-mean processes and the limitation is stated. The known N1 AC exceedance (7.0%, upper CI 8.3%) is reported for every test; it is a limitation, not a finding.

6.5 P5, P5FA: preprocessing with standard tests (step 5)

Item	Specification
Question / hypothesis	Q2 and the head-to-head “fix the data vs fix the test”. No directional hypothesis.
Model, groups	P5: ou_csd, P1 design. P5FA: stable, N2 design. Arms regular_N, informative_reduced_N. All 13 frozen E1 representations. Tests ar1_rep, shuffle_rep.
n, seeds	P5 ids 0–999 (pairs with E1 and P1); P5FA ids 0–1999 (pairs with E3 and N2); seed base 3,000,000.
Outcomes	Recovery det(inf, r, n) − det(inf, index, n); gap det(inf, r, n) − det(reg, index, ou_ss); inf − reg within r; FA(r, n) with TOST. Plus the head-to-head det(inf, r, ar1_rep) − det(inf, index, ou_ss).
CIs, multiplicity	Family S-Q2: 95% unadjusted CIs, Holm across the 13 representations × 2 tests × 2 indicators.
Interpretation	Every recovery estimate is classified with the four-way rule (Section 5b) using P5FA for the same representation, test, arm and indicator: NO_RECOVERY, IMPROVEMENT_FA_CONTROL_NOT_ESTABLISHED, GENUINE_RECOVERY or IMPROVEMENT_AT_FALSE_ALARM_COST. **Only GENUINE_RECOVERY supports a claim that preprocessing recovers detection**; an improvement without established false-alarm control is never called recovery, and “at the cost of excess false alarms” is used only when the false-alarm rate is demonstrably above 7.5%. If a representation with ar1_rep reaches GENUINE_RECOVERY and its head-to-head with index + ou_ss has a 95% CI inside ±3 pp, the message becomes “fix the data or fix the test”. The same rule is applied to the E1/E3 (ou_ss) results.

6.6 FOLDP, FOLD2: tipping-point model with adequate power (step 6)

Item	Specification
Question / hypothesis	Does Q1 (and Q3) hold for a bifurcation-type model?
Pilot (selection only)	Grid $\sigma_f \in \{0.05, 0.07, 0.10\} \times r_{end} \in \{0.05, 0.10\}$; 300 patients per point, seed base 8,000,000; regular_N arm only, ou_ss, index. Eligible: tipping ≤ 5%. Choose the eligible point with regular_N variance detection closest to 50% and inside 35–65%; ties → smaller σ_f, then larger r_{end}. The informative arm is never computed in the pilot. No eligible point → FOLD2 is not run and the fold model is reported as uninformative.
Main run	FOLD2: selected setting, step, m = 2, primary trigger; four arms; three tests; 1,400 patients, seed base 7,000,000 (expected ≥ 1,330 non-tipped). Tipped patients counted and excluded as frozen.
Power	With discordance about 0.40, paired 95% CI half-width ≈ $1.96 \cdot \sqrt{(0.40/1330)} = 3.4$ pp; power to detect an 8 pp difference > 0.99, a 5 pp difference about 0.93.
Interpretation	Frozen ±5 pp categories. Results are conditional on non-tipping, stated as such.

6.7 E2, E2FA: continuous-time method (step 2)

Item	Specification
Question / hypothesis	Q2: does a model-based continuous-time estimator (window-wise OU fit) recover detection, and at what false-alarm rate?
Method (frozen)	ct|W30: 43 windows of width 30 (starts j·70/42), ≥ 8 observations per window; in each window fit the stationary OU + measurement-error model by exact Kalman likelihood (same grid as ou_ss, μ = window mean, V profiled); indicators: fitted latent variance V and $e^{-\theta}$; Kendall tau across windows; ou_ss test; grid-edge estimates counted and reported. No tuning.
n, seeds	E2 exactly as frozen: ou_csd, ids 0–499. E2FA: stable, ids 0–999 (**n = 1,000 is the target**), seed base 3,000,000. Deviation (recorded): frozen E3 lists ct for 2,000 patients (about 69 million window fits). At n = 1,000 the TOST passes for observed 3.4–6.1% (probability 0.94 at a true 5%). Only if the timing pilot projects > 12 hours may n be reduced to 500, and only with recorded approval before the run. An n = 500 result is **strictly descriptive**: it cannot establish CALIBRATED status and cannot support recovery or any controlled-false-alarm claim.
Interpretation	Classified with the four-way rule (Section 5b). GENUINE_RECOVERY requires E2FA at n = 1,000 with ct|W30 CALIBRATED (or LOW) in the matching arm and indicator. E2 has 500 patients, so its paired CI half-width is about ±5.5 pp; stated.

7. What each outcome allows the paper to say

Result	Claim permitted	Claim not permitted
G1a fails	Q1 and Q2 findings only; timing-aware test described as uncalibrated in the slow regime.	Any recommendation of ou_ss.
G1a passes, H1 and H2 supported, ou_ss AND comparator CALIBRATED (G1a and XN2)	"Timestamp-blind surrogate tests lose variance detection under irregular sampling; a timing-aware test recovers it at controlled false-alarm rates." Scope from G1b–d, XS2.	That ou_ss rescues AC (unless F1 shows it); that the effect is state-dependent (unless 6.3 rule met).
H1/H2 supported but comparator LOW	"ou_ss detects more than a conservative timestamp-blind test", judged on the size-adjusted analysis.	"At controlled false-alarm rates" for the raw comparison.
H1/H2 supported but comparator HIGH_NOT_EXCLUDED or HIGH	Only what the size-adjusted analysis supports; comparator's false-alarm control described as not established (or, for HIGH, as demonstrably excessive).	Any superiority claim from the raw comparison.
H1 only	"ou_ss is more powerful in this setting."	That the gain is specific to irregular sampling.
P5/E1/E2: GENUINE_RECOVERY	"Regularising the data with representation r recovers detection" (at calibrated, or conservative, false-alarm rates).	General claims about other representations.
IMPROVEMENT_FA_CONTROL_NOT_ESTABLISHED	"Detection improvement with false-alarm control not established."	"Recovers detection"; "at the cost of excess false alarms".
IMPROVEMENT_AT_FALSE_ALARM_COST	"Detection improvement at the cost of excess false alarms."	"Recovers detection".
No representation recovers	"Common preprocessing does not recover the loss." (already seen for binned|60|linear_interp with ou_ss)	—
XS1 escalation component ≈ P1/F1	"A rise in sampling density".	"State-dependent timing".

8. Workload (estimate, to be verified)

These are estimates, not verified runtimes. Measured rates on the production computer: 0.098 s per patient × arm × analysis × test unit in P1, 0.078 in E1 and 0.077 in E3 (200 surrogates each). Stage 0 runs every block on 5 patients (outputs discarded) and reports a projected runtime for approval before any production run.

Blocks	Units	Estimated hours
G1a–G1d	72,000	1.5–2.0
F1	24,000	0.5–0.7
XS1, XR1–XR10, XS2, XN1, XN2	177,000	3.8–4.8
P5, P5FA	156,000	3.3–4.2
FOLDP, FOLD2	18,600	0.4–0.5
E2, E2FA (continuous-time)	—	unknown: each surrogate refits 43 windows; measured in Stage 0
Total excluding E2/E2FA	**447,600**	**about 9.6–12.2**

Runs are split into stages (the run_order list in the JSON) so each can run overnight and be checked before the next.

9. Reproducibility audit and verification (every block)

- **Frozen inputs:** A1 JSON hash checked by the code at start (like the Phase 2 config); parent and grandparent hashes checked; code SHA-256 of every module recorded in each manifest.
- **Seeds and streams:** seed base and patient ids per block as listed; every stream name planned and checked unique before the run; a run that uses an unplanned stream aborts with no record.
- **Outputs:** outputs_addendum_A1/ only. Per block: patient-level rows CSV (every patient × arm × analysis × test, tau and p for both indicators, evaluability, counts, escalation time), manifest (command, software versions, platform, hashes), execution log, and an immutable completion record written last.
- **Identity checks:** before results are summarised, every original-seed block must reproduce the completed Phase 2 rows exactly (tau and p, both indicators) for every arm/analysis/test already run, and P5 must

reproduce E1's regular/informative series. Any mismatch is an implementation error: stop and report, no results used.

- **Code checks before production:** unit tests for every new component (stable_slow stationarity, poisson_matched count and uniformity, ar1_rep/shuffle_rep on representations, fold_v2 grid and pilot rule, A1 hash lock); plumbing run (discarded); independent cold code review.
- **Independent numerical verification:** after each stage, the summary is recomputed from the patient-level rows by a separate script written independently of the summariser.
- **Deviations:** append-only deviations.json, one dated entry per departure from this protocol, written before the affected run. The E2FA size reduction is entry 1.
- **Never overwritten:** completed Phase 2 outputs, the earlier E1/E3 outputs, any completion record.

10. Approval and freeze

- Version history: v1.0 proposed; v1.1 revised (three corrections); v1.2 revised (HIGH wording); approved for freezing on 30 Oct 2026.
- Frozen machine-readable file: Simulation_Addendum_A1_v1.2_FROZEN.json, SHA-256 07e896f9bc70eb1356efbfcbfce08b0e6e06d470c28485f5ed592d66d0bffe4b. The A1 code hard-codes this hash and refuses to run if the file differs. Any later change needs a new version and a dated deviations.json entry.
- Stage 0 (tests, plumbing, identity checks, timing pilot) runs first; its projected runtime is shown for approval before Stage 1.

11. Open questions answered by experiments, and what needs no new run

Open question	Addressed by
Mechanism (1/m compression unsupported)	No run: replaced by the closed-form account (lag-1 AC under exponential gaps $\lambda/(\lambda+\theta)$ vs $e^{-\theta/\lambda}$ under even spacing; effective sample size) plus tau-scale results from F1 and X-blocks.
Bifurcation model	FOLDP/FOLD2 (6.6).
Alternatives (binning, interpolation, fixed-time, model-based)	E1/E3 (running), P5/P5FA, E2/E2FA. Elapsed-time analysis is an identity by construction and is not counted as evidence (no run).
Ramp-shaped intensity	XR9 (frozen ramp, capped at 2×). A ramp matched to the observed MIMIC-IV profile needs the step-7 data first; specified in the MIMIC plan.
Equivalence for false alarms	Already addressed (Phase 1 TOST); extended by G1, XN1, XN2, P5FA, E2FA.
Dose-response	XR5–XR8 at n = 1,000 with a paired trend estimate (6.3).
Other points	No simulation: purpose and audience, MIMIC-IV plan, literature, removal of legacy results, figures, AI declaration.